MILITARY SPECIFICATION

FUZE, PD, M525A1E1, HEAD ASSEMBLY FOR

This Amendment forms a part of Military Specification MIL–C–46220A (Ord) dated 20 April 1962.

Page 1 ff, suffix to specification symbol. Change "MIL–F–46220A (Ord)" to "MIL–F–46220A (MU)" wherever it appears throughout the specification.

Page 2, Drawings. Add "8799125—Fuze, PD, M525A1, Assembly."

Page 2, Publications, Ordnance Corps. Delete "IEL–8815821" and substitute "IEL–8799127."

Page 3, paragraph 3.3.6, line 2. Delete "1.8" and substitute "2.0."

Page 4, paragraph 3.9, line 4. Insert the word "cracks" after the word "burrs."

Page 4: Add paragraph 3.10 as follows:

"3.10 Ballistic performance.

3.10.1 *Nonfunctioning.* The head assembly shall not function when tested as specified in 4.4.15.1.

3.10.2 *Functioning.* There shall be no evidence of a premature burst and the head assembly shall cause the shell to function "high order" when tested as specified in 4.4.15.2. (A premature burst is defined as functioning of the shell in flight before impact.)"

Page 6, paragraph 4.2. Delete and substitute:

"4.2 Initial production inspection.

4.2.1 *Examination and static tests.*

4.2.1.1 *Submission.* The contractor shall submit 300 consecutively produced complete head assemblies which have been produced by the production process which the contractor will use in fulfilling the contract to a Government approved facility designated by the contracting officer. In addition, 25 complete sets of parts (i.e., 25 each of every individual component and every subassembly) shall be submitted except that quantities for the following shall be—

175	train pinion-wheels
50	each of lower bushings, escape wheels, pallets, gears, striker assemblies and upper plate assemblies
5	heads from each mold cavity
5	strikers from each mold cavity

The initial production quantity shall be accompanied by certificates of analysis for all materials. An initial production quantity shall also be submitted whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government. Of the 300 head assemblies, 50 will be used for testing as specified in 4.2.1.2 and, in the event the entire initial production quantity passes the inspections specified, the remaining 250 head assemblies will be retained by the engineering agency as a control sample.

4.2.1.2 *Inspections to be performed.* Each assembly, subassembly and component part shall be inspected for all of the inspections specified in 4.3.2 and 4.3.3 of this specification and for all requirements of the applicable drawings. The 50 head assemblies shall be subjected to the following tests in the order specified except that only 12 of the assemblies shall be subjected to the jolt and jumble tests.

(a) Transportation vibration (see 4.4.2)
(b) Nonpositioning (see 4.4.3)
(c) Arming (see 4.4.5)
(d) Arming delay (see 4.4.6)

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(e) Nonarming (see 4.4.1)
(f) Positioning (see 4.4.4)
(g) Jolt and jumble (see 4.4.7)

4.2.2 *Ballistic tests.*

4.2.2.1 *Submission.* The contractor shall submit 1000 consecutively produced complete head assemblies, which have been produced by the same production process as those submitted in accordance with 4.2.1.1 to a Government proving ground designated by the contracting officer.

4.2.2.2 *Tests to be performed.* The head assemblies shall be tested in accordance with 4.4.15.1 or 4.4.15.2, as applicable, except that quantities and test conditions shall be as specified in table I.

TABLE I

Requirement	Quantity	Temperature	Charge
Nonfunctioning (see 3.10.1)	250	70° F.	0
Functioning (see 3.10.2)	125	125° F.	0
Functioning	125	125° F.	6
Functioning	125	70° F.	0
Functioning	125	70° F.	6
Functioning	125	minus 40° F.	0
Functioning	125	minus 40° F.	6

4.2.3 *Rejection.* If any assembly, subassembly or component part fails to comply with the requirements for the inspections performed in accordance with 4.2.1.2 the initial production quantity shall be rejected. In the ballistic test, if any round functions in the nonfunctioning phase, or if more than 5 rounds fail to function in any group of 125 fired for functioning, or if more than 25 rounds fail in the total of 750 fired in the functioning phase, the preproduction quantity shall be rejected. In the event of rejection the Government reserves the right to require the contractor to take corrective action and submit new or supplemental quantities until such time as an acceptable sample is submitted or until otherwise directed by the contracting officer.

Page 13, paragraph 4.3.2.11, Major category:

Delete "AQL 0.40 percent" and substitute "AQL 0.65 percent"
Delete defect 102
Add the following defects:

"104. Distance from face of flange to center of small holes ..... Gage
105. Location of small holes with respect to lugs ..... Gage
106. Diameter of small holes, min. ..... Gage"

Page 20, paragraph 4.3.3.14. Delete this paragraph in its entirety.

Page 21. Add paragraph 4.3.3.18 as follows:

"4.3.3.18 *Ballistic tests.* The following tests shall be performed as specified in 4.4.15.

4.3.3.18.1 Nonfunctioning (see 3.10.1) Critical defect.

4.3.3.18.1.1 Beginning with the first production lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified herein, a sample of 25 head assemblies shall be selected from each lot for this test. If any assembly fails to comply with the applicable requirement, the lot shall be rejected.

4.3.3.18.1.2 After three consecutive lots have met the criteria of 4.3.3.18.1.1, a sample of 10 head assemblies shall be selected from each lot for this test. If any assembly fails to comply with the applicable requirements the lot shall be rejected.

4.3.3.18.2 Functioning (see 3.10.2) Major defect.

4.3.3.18.2.1 Beginning with the first production lot produced and continuing until

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three consecutive lots have complied with the acceptance criteria specified herein, a sample of 100 head assemblies shall be selected for this test. If 4 or more assemblies fail to comply with the applicable requirement, the lot shall be rejected.

4.3.3.18.2.2 After three consecutive lots have met the criteria of 4.3.3.18.2.1, a sample of 40 head assemblies shall be selected from each lot for this test. If 4 or more assemblies fail to comply with the applicable requirement, the lot shall be rejected. If 2 or 3 assemblies fail, a second sample of 80 assemblies shall be tested. If the combined number of failures in both the first and second samples is 4 or more, the lot shall be rejected."

Page 21. Add paragraph 4.3.3.19 as follows:

"4.3.3.19 *Control sample.* A control sample of 250 head assemblies shall be selected at random from each lot and be forwarded to the responsible engineering agency for such inspection and evaluation as is deemed appropriate. Results of this evaluation will not affect lot acceptance but may form the basis for corrective action in subsequent lots."

Page 21, paragraph 4.3.4. Delete and substitute—

"4.3.4 *Inspection equipment.* Index of Inspection Equipment Lists Number IEL 8799127 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.2. Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL–I–45607."

Page 22, paragraph 4.4.4, line 7. Delete "dented" and substitute "detented."

Page 22, paragraph 4.4.4. Add the following:

"*Note.* For the performance of this test on the initial production sample (see 4.2.1.2) reassembly of the pull wire is not required."

Page 24. Add paragraph 4.4.15 as follows:

"4.4.15 *Ballistic performance.*

4.4.15.1 *Nonfunctioning.* The sample head assemblies shall be assembled into complete Fuzes as shown on dwg. 8799125 using an approved lot of M52A2 PD Fuzes (see 6.8). The fuzes shall then be assembled to 81MM M43A1 HE complete round cartridges with a propellant charge consisting of an ignition cartridge only (charge 0) and fired horizontally at a ¼ inch thick plywood target placed 200 feet from the muzzle of the mortar tube. Observation shall be made for functioning on the target.

4.4.15.2 *Functioning.* The sample head assemblies shall be assembled into complete Fuzes as shown on dwg. 8799125 using an approved lot of M52A2 PD Fuzes (see 6.8). The fuzes shall then be assembled to 81MM M43A1 HE complete round cartridges and fired at an elevation of 45 degrees or greater for impact on firm ground. For the initial production ballistic firing, sample quantities shall be as specified in 4.2.2, table I, and applicable conditioning shall be for 72 hours, minimum. For tests of initial production and production lots, one half of the sample quantity shall be fired at ambient temperature with ignition cartridges only (charge 0) and the remaining sample units shall be fired at ambient temperature with six propellant increments (charge 6). Observation shall be made for premature bursts and for functioning upon ground impact."

Page 26, paragraph 6.1. Delete and substitute—

"**6.1 Ordering data.** Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Subject to rights in data acquired under the contract, engineering drawings and lists provided and revised under this specification may be used by the Department of Defense activities for design, procurements, manufacture, testing evaluation, production, production and receiving inspection, overhaul shipping, storage, identification of stock, ordering and

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storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey, and whenever engineering drawings are needed."

Page 26, paragraph 6.2. Delete and substitute—

"6.2 Inspection equipment. The contractor shall design inspection equipment as required by the referenced Inspection Equipment List in accordance with the instructions of paragraphs 6.2.1 through 6.2.5.

6.2.1 Inspection Equipment Lists (IEL) indicate the availability of inspection equipment design by showing in the "number" column of the list of inspection equipment—00 Form 1242–3—the numbers of drawings of existing equipment designs, or codes indicating "commercial item" or "contractor design". Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.2.2 and 6.2.3. Action required of the contractor with respect to "commercial item" inspection equipment, is described in paragraph 6.2.4. The contractor will be required to prepare drawings for all equipment coded as "contractor design" in the number column. The code will further indicate whether detailed or specification control drawings are required. These contractor designs must be approved by the Government prior to the fabrication or procuring of the equipment. Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND.

6.2.2 *Detailed drawings.* Detailed drawings are drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. Contractors need provide no design when a detailed drawing is listed for an item of inspection equipment and the contractor elects to use the Government design. The contractor may, however, with the approval of the Government, design alternate inspection equipment to facilitate his operations. Such contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND.

6.2.3 *Specification control drawings.* Specification control drawings depict the device in outline, descriptive, diagrammatic, or pictorial form only and specify the required performance or other characteristics. Contractors must prepare drawings of their designs in support of specification control drawings. The IEL code will indicate whether the drawings must be detailed. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SNUPA–ND.

6.2.4 *Commercial items.* Commercial item inspection equipment is equipment that has a universal application for a specific function. It is comprised of items commonly used by industry and government. Contractors are not required to furnish drawings of commercial item inspection equipment, but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.2.5 *Use of standard measuring equipment.* If the contractor desires to use standard measuring equipment in lieu of government or contractor designed gages, diagrammatic sketches or other descriptions of the use of such equipment shall be submitted to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND for approval."

Page 27. Add paragraph 6.8 as follows:

"6.8 Approval of M52A2 PD Fuze for use in ballistic tests. Approval of M52A2 PD Fuzes for use in the ballistic tests prescribed herein shall be determined by selecting a random sample of 300 from a lot of 5000 or

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more and, after assembly to M43A1 HE complete round cartridges, firing at an angle of 45 degrees elevation or greater for impact on firm ground. One hundred fifty (150) of the sample units shall be fired with ignition cartridges only (charge 0) and 150 sample units shall be fired with 6 propellant increments (charge 6). If there are no premature bursts and if no more than 2 duds occur, the lot shall be approved for use."

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